

Fixing the Element Arrays

Problem Description

Ilya works in a magic laboratory, Today she gets a challenging task:

Due to the carelessness of other staff members, the element materials used to maintain the element arrays have been placed in the wrong positions. The positions for fire element materials and water element materials have been swapped:

- The water element array position has n fire element materials stacked in a pile from top to bottom,
- The fire element array position has n water element materials stacked in a pile from top to bottom.
- Each element material has its own magic power x_i . The water element materials, from bottom to top, have magic powers $1, 2, 3, \dots, n$, and the fire element materials are the same.

However, these element materials are very unstable and must be moved under the following conditions:

- For safety reasons, Ilya can only move one magic material at a time;
- To ensure the stability of the elements, no element material can be placed on top of another element material with higher magic power, but it can be placed on top of one with lower or equal magic power; different element materials can also directly touch and stack on each other;
- Element materials can only be placed on element arrays and stacked in piles from top to bottom.

Fortunately, there's an empty grass element array nearby that can be used to temporarily store element materials, but the same movement conditions still apply.

Since moving element materials is tiring each time, Ilya wants to know the minimum number of moves needed to return all element materials to their correct positions.

The answer might be very large, so please take the modulo of 1000001011 (which is a prime number).

Input and Output

Input Format

The input is a single line containing an integer n , representing the number of element materials in both the water element pile and the fire element pile.

Output Format

Output a single integer, representing the minimum number of moves required.

Data Range

- For 30% of the data, $n \leq 5$;
- For 80% of the data, $n \leq 10^7$;
- For 100% of the data, $n \leq 10^{12}$;

Example Test Cases

Example Input 1:

3

Example Output 1:

25

Example Input 2:

1234567890

Example Output 2:

612318234